

Side effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness
- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- * Pain at injection site
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose

- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Especially at initiation of treatment or when dose increases
- Tolerance often develops over time

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Take largest dose at bedtime to avoid sedative effects during the day
- Switch to another agent
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- Oral: 4–40 mg/day in divided doses
- Intravenous (adults): 5 mg/minute
- Intravenous (children): 0.25 mg/kg every 3 minutes

Dosage Forms

- Tablet 2 mg, 5 mg, 10 mg
- Nasal 5 mg/spray, 7.5 mg/spray, 20 mg/spray
- Concentrate 5 mg/mL
- Solution 5 mg/5 mL
- Injection vial 5 mg/mL, 10 mg/2 mL, 50 mg/10 mL
- Rectal gel 2.5 mg/0.5 mL, 10 mg/2 mL, 20 mg/4 mL

How to Dose

- Oral (anxiety, muscle spasm, seizure): 2–10 mg, 2–4 times/day
- Oral (alcohol withdrawal): initial 10 mg, 3–4 times/day for 1 day; reduce to 5 mg, 3–4 times/day; continue treatment as needed
- Liquid formulation should be mixed with water or fruit juice, applesauce, or pudding
- Because of risk of respiratory depression, rectal diazepam treatment should not be given more than once in 5 days or more than twice

during a treatment course, especially for alcohol withdrawal or status epilepticus

Dosing Tips

- * Only benzodiazepine with a formulation specifically for rectal administration
- * One of the few benzodiazepines available in an oral liquid formulation
- * One of the few benzodiazepines available in an injectable formulation
- Diazepam injection is intended for acute use; patients who require long-term treatment should be switched to the oral formulation
- Use lowest possible effective dose for the shortest possible period of time (a benzodiazepine-sparing strategy)
- Assess need for continued treatment regularly
- Risk of dependence may increase with dose and duration of treatment
- For interdose symptoms of anxiety, can either increase dose or maintain same total daily dose but divide into more frequent doses
- Can also use an as-needed occasional “top-up” dose for interdose anxiety
- Because some anxiety disorder patients and muscle spasm patients can require doses higher than 40 mg/day or more, the risk of

dependence may be greater in these patients

- Frequency of dosing in practice is often greater than predicted from half-life, as duration of biological activity is often shorter than pharmacokinetic terminal half-life

Overdose

- Fatalities can occur; hypotension, tiredness, ataxia, confusion, coma

Long-Term Use

- Evidence of efficacy up to 16 weeks
- Risk of dependence, particularly for treatment periods longer than 12 weeks and especially in patients with past or current polysubstance abuse
- Not recommended for long-term treatment of seizure disorders

Habit Forming

- Diazepam is a Schedule IV drug
- Patients may develop dependence and/or tolerance with long-term use

How to Stop

- Patients with history of seizure may seize upon withdrawal, especially if withdrawal is abrupt

- Taper by 2 mg every 3 days to reduce chances of withdrawal effects
- For difficult-to-taper cases, consider reducing dose much more slowly after reaching 20 mg/day, perhaps by as little as 0.5–1 mg every week or less
- For other patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization
- Be sure to differentiate reemergence of symptoms requiring reinstitution of treatment from withdrawal symptoms
- Benzodiazepine-dependent anxiety patients and insulin-dependent diabetics are not addicted to their medications. When benzodiazepine-dependent patients stop their medication, disease symptoms can reemerge, disease symptoms can worsen (rebound), and/or withdrawal symptoms can emerge

Pharmacokinetics

- Elimination half-life 20–50 hours
- Substrate for CYP450 2C19 and 3A4
- Food does not affect absorption

Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)
- Cimetidine may reduce the clearance and raise the levels of diazepam
- Flumazenil (used to reverse the effects of benzodiazepines) may precipitate seizures and should not be used in patients treated for seizure disorders with diazepam

Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Dosage changes should be made in collaboration with prescriber
- Use with caution in patients with pulmonary disease; rare reports of death after initiation of benzodiazepines in patients with severe

pulmonary impairment

- History of drug or alcohol abuse often creates greater risk for dependency
- Some depressed patients may experience a worsening of suicidal ideation
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)

Do Not Use

- If angle-closure glaucoma
- If there is a proven allergy to diazepam or any benzodiazepine

Special populations

Renal Impairment

- Initial 2–2.5 mg, 1–2 times/day; increase gradually as needed

Hepatic Impairment

- Initial 2–2.5 mg, 1–2 times/day; increase gradually as needed

Cardiac Impairment

- Benzodiazepines have been used to treat anxiety associated with acute myocardial infarction

- Diazepam may be used as an adjunct during cardiovascular emergencies

Elderly

- Initial 2–2.5 mg, 1–2 times/day; increase gradually as needed



Children and Adolescents

- 6 months and up: initial 1–2.5 mg, 3–4 times/day; increase gradually as needed
- Parenteral: 30 days or older
- Rectal: 2 years or older
- Long-term effects of diazepam in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Possible increased risk of birth defects when benzodiazepines are taken during pregnancy

- Because of the potential risks, diazepam is not generally recommended as treatment for anxiety during pregnancy, especially during first trimester
- Drug should be tapered if discontinued
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy
- Seizures, even mild seizures, may cause harm to the embryo/fetus

Breast Feeding

- Unknown if diazepam is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects of benzodiazepines on nursing infants have been reported and include feeding difficulties, sedation, and weight loss

The art of psychopharmacology

Potential Advantages

- Rapid onset of action
- Availability of oral liquid, rectal, and injectable dosage formulations

Potential Disadvantages

- Euphoria may lead to abuse
- Abuse especially risky in past or present substance abusers
- Can be sedating at doses necessary to treat moderately severe anxiety disorders

Primary Target Symptoms

- Panic attacks
- Anxiety
- Incidence of seizures (adjunct)
- Muscle spasms

Pearls

- Can be a useful adjunct to SSRIs and SNRIs in the treatment of numerous anxiety disorders, but not used as frequently as other benzodiazepines for this purpose
 - Not effective for treating psychosis as a monotherapy, but can be used as an adjunct to antipsychotics
 - Not effective for treating bipolar disorder as a monotherapy, but can be used as an adjunct to mood stabilizers and antipsychotics
- * Diazepam is often the first-choice benzodiazepine to treat status epilepticus, and is administered either intravenously or rectally

- Because diazepam suppresses stage 4 sleep, it may prevent night terrors in adults
- May both cause depression and treat depression in different patients
- Was once one of the most commonly prescribed drugs in the world and the most commonly prescribed benzodiazepine
- * Remains a popular benzodiazepine for treating muscle spasms
- A commonly used benzodiazepine to treat sleep disorders
- * Remains a popular benzodiazepine to treat acute alcohol withdrawal
- Not especially useful as an oral anticonvulsant
- * Multiple dosage formulations (oral tablet, oral liquid, rectal gel, injectable) allow more flexibility of administration compared to most other benzodiazepines
- When using to treat insomnia, remember that insomnia may be a symptom of some other primary disorder itself, and thus warrant evaluation for comorbid psychiatric and/or medical conditions
- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment



Suggested Reading

Ashton H . Guidelines for the rational use of benzodiazepines. When and what to use . *Drugs* 1994 ;**48** :25 –40 .

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De Negri M , Baglietto MG . Treatment of status epilepticus in children . *Paediatr Drugs* 2001 ;**3** :411–20 .

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Mandelli M , Tognoni G , Garattini S . Clinical pharmacokinetics of diazepam . *Clin Pharmacokinet* 1978 ;**3** :72 –91 .

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Rey E , Treluver JM , Pons G . Pharmacokinetic optimization of benzodiazepine therapy for acute seizures. Focus on delivery routes . *Clin Pharmacokinet* 1999 ;**36** :409–24 .

Diphenhydramine

Therapeutics

Brands

- BenadrylSominex
- *see index for additional brand names*

Generic?

Yes

Class

- Antihistamine; anticholinergic agent

Commonly Prescribed for

(bold for FDA approved)

- **Allergy symptoms**
- **Motion sickness**
- **Occasional sleeplessness**
- **Antiparkinsonism**
- Drug-induced parkinsonism

 **How the Drug Works (for drug-induced parkinsonism)**

- Diminishes the excess acetylcholine activity caused by removal of dopamine inhibition when dopamine receptors are blocked
- Also has potent histamine 1 antagonist properties

How Long Until It Works

- For drug-induced parkinsonism and in Parkinson's disease, onset of action can be within minutes or hours

If It Works (for drug-induced parkinsonism)

- Reduces motor side effects
- Does not lessen the ability of antipsychotics to cause tardive dyskinesia

If It Doesn't Work (for drug-induced parkinsonism)

- Consider switching to trihexyphenidyl, benztropine, or a benzodiazepine
- Disorders that develop after prolonged antipsychotic use may not respond to treatment
- Consider discontinuing the agent that precipitated the parkinsonism



Best Augmenting Combos for Partial Response or Treatment Resistance

- If ineffective, switch to another agent rather than augment
- Diphenhydramine itself is an augmenting agent to antipsychotics

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Blocking histamine 1 receptors can cause sedation
- Preventing the action of acetylcholine on muscarinic receptors can cause anticholinergic effects such as dry mouth, blurred vision, constipation

Notable Side Effects

- Sedation, dizziness
- Constipation, nausea
- Dry mouth, blurred vision



Life-Threatening or Dangerous Side Effects

- Rare convulsions (at high doses)
- Urinary retention
- Tachycardia, cardiac arrhythmias
- Confusion
- Paralytic ileus/bowel obstruction

Weight Gain



- Frequent and can be significant in amount

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- For confusion or hallucinations, discontinue use
- For sedation, lower the dose and/or take the entire dose at night
- For dry mouth, chew gum or drink water
- For urinary retention, obtain a urological evaluation; may need to discontinue use

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- Oral: 50 mg/day
- Injection: 10–50 mg

Dosage Forms

- Injection 50 mg/mL
- Capsule 50 mg
- Elixir 12.5 mg/5 mL
- Also available in formulations in combination with other medications

How to Dose

- Injection: 10–50 mg intravenously at a rate not exceeding 25 mg/min or deep intramuscularly; can dose at 100 mg if required; maximum daily dose 400 mg



Dosing Tips

- If drug-induced parkinsonism occurs soon after initiation of a neuroleptic drug, it is likely to be transient; thus, attempt to withdraw diphenhydramine after 1–2 weeks to determine if still needed
- The injection should be used for parkinsonism only if oral therapy is impossible or contraindicated

Overdose

- CNS depression, CNS stimulation (more likely in pediatric patients), dry mouth, dilated pupils, flushing, gastrointestinal symptoms

Long-Term Use

- Safe
- Effectiveness may decrease over time, even after a few doses, but side effects such as cognitive impairment and sedation may persist

Habit Forming

- No

How to Stop

- Tapering generally not necessary

Pharmacokinetics

- Plasma half-life approximately 8 hours; may be longer in children and in the elderly



Drug Interactions

- May potentiate the effects of other CNS depressants
- If anticholinergic agents are used with diphenhydramine, the anticholinergic effects may be enhanced



Other Warnings/Precautions

- Use with caution in patients with a history of bronchial asthma, lower respiratory disease, increased intraocular pressure, hyperthyroidism, cardiovascular disease, or hypertension

- May have additive effects if taken with anticholinergic agents

Do Not Use

- If patient is breast feeding
- In neonates or premature infants
- In patients with glaucoma, particularly angle-closure glaucoma
- In patients with pyloric or duodenal obstruction, stenosing peptic ulcers, prostate hypertrophy, or bladder neck obstructions
- If patient is taking an MAOI
- If there is a proven allergy to diphenhydramine

Special populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- No dose adjustment necessary

Cardiac Impairment

- Not systematically evaluated in patients with cardiac impairment

Elderly

- Some patients may tolerate lower doses better
- Diphenhydramine injection is preferred for parkinsonism in the elderly who are unable to tolerate more potent agents

Children and Adolescents

- Not recommended for children under age 12
- Injection: 5 mg/kg/24 hours or 150 mg/m² /24 hours; maximum daily dose 300 mg; divide into 4 doses; can be given intravenously at a rate not exceeding 25 mg/min, or deep intramuscularly
- Contraindicated in neonates and premature infants

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Animal studies have not shown adverse effects

Breast Feeding

- Contraindicated

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Potential Advantages

- Mild cases of parkinsonism

Potential Disadvantages

- Can be too sedating for some patients
- Can cause confusion in elderly patients and in patients with dementia

Primary Target Symptoms

- Drug-induced parkinsonism



Pearls

- Can be useful for occasional insomnia
- Patients with cognitive impairment may do poorly
- Can cause cognitive side effects with chronic use, so periodic trials of discontinuation may be useful to justify continuous use, especially in institutional settings when used as an adjunct to antipsychotics
- Can be abused in institutional or correctional settings



Suggested Reading

Gonzalez F. Diphenhydramine may be useful as a palliative treatment for patients dying with Parkinson's disease and tremors: a case report and discussion . *Am J Hosp Palliat Care* 2010 ;**26** (6):474 –5 .

Disulfiram

Therapeutics

Brands

- Antabuse
- *see index for additional brand names*

Generic?

Yes

Class

- Alcohol dependence treatment

Commonly Prescribed for

(bold for FDA approved)

- **Maintenance of alcohol abstinence**

How the Drug Works

- Irreversibly inhibits aldehyde dehydrogenase, the enzyme involved in second-stage metabolism of alcohol
- Alcohol is metabolized to acetaldehyde, which in turn is metabolized by aldehyde dehydrogenase; thus, disulfiram blocks this second-stage metabolism

- If alcohol is consumed by a patient taking disulfiram, toxic levels of acetaldehyde build up, causing unpleasant side effects
- This aversive experience ideally leads to negative conditioning, in which patients abstain from alcohol in order to avoid the unpleasant effects

How Long Until It Works

- Disulfiram's effects are immediate; patients should not take disulfiram until at least 12 hours after drinking

If It Works

- Increases abstinence from alcohol

If It Doesn't Work

- Patients who drink alcohol while taking disulfiram can experience side effects, including alcohol toxicity
- Evaluate for and address contributing factors
- Consider switching to another agent



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation with behavioral, educational, and/or supportive therapy in groups or as an individual is probably key to successful treatment